iQuCodeFest 2026
Quantum Images

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Sherbrooke and Calgary

Tania Belabbas
tania.belabbas@usherbrooke.ca

Quantum Image

Welcome!

- **28 hours hackathon.**
- Points to gain from various challenges.
- **Mentors in the room with you.**
- **3 judges**.

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Welcome!

- **Main challenge**.
- **Side-quest** challenges.
- International Year of Quantum **Quiz**.

Main challenge

Quantum Image

Quantum Image

Main challenge

- **Goal of the challenge** : Explore whether quantum computing can contribute meaningfully to image workflows through two tracks :
  - Quantum image analysis

Quantum Image

Main challenge

- **Goal of the challenge** : Explore whether quantum computing can contribute meaningfully to image workflows through two tracks :

  - Quantum image analysis
    - Detect, classify, segment, reconstruct, compare, or enhance images using hybrid quantum-classical methods

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Main challenge

- **Goal of the challenge** : Explore whether quantum computing can contribute meaningfully to image workflows through two tracks :

  - Quantum image analysis
  - Quantum image generation

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Quantum Image

Main challenge

- **Goal of the challenge** : Explore whether quantum computing can contribute meaningfully to image workflows through two tracks :
  - Quantum image analysis
  - Quantum image generation
    - Create or synthesize images using quantum or hybrid methods

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Main challenge

- **Goal of the challenge** : Explore whether quantum computing can contribute meaningfully to image workflows through two tracks :
  - Quantum image analysis
  - Quantum image generation

Identify a problem where quantum ideas are **pertinent**, **justify** the choice of solution, and build a concrete **proof of concept.**

Refer to the rule book for details on what is expected.

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Main challenge

- Before the end of the first day, you must :

Add the mentors as collaborators in the GitHub repository that will contain your solution

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Main challenge

- By the end of the hackathon [3:15PM Sherbrooke - 1:15PM Calgary], your repository must contain :
  - A notebook, with outputs, showcasing your solution
  - Your well document python code
  - A presentation detailing your solution [PDF, Powerpoint or Keynote]
    - This will be used as visual support for your presentation

Quantum Image

Main challenge

- By the end of the hackathon [3:15PM Sherbrooke - 1:15PM Calgary], your repository must contain :
  - **Add a requirement.txt**
  - Make sure to present your solution as clearly as possible
  - Verify your solution/code with a mentor before the end of the hackathon

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Side quests

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Side-quest challenges

- **Four** introductory notebooks. Get yourself familiar with **Qiskit** and basic quantum programming.

- A markdown guide document **introducing good practices** with Python environment, Github, and more.

- **Three side quests** that can earn you more points and guide you through the main challenge. Spend your time strategically.

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Side-quest challenges

- **Get your side quests graded by a mentor before the end of the hackathon to get the points associated to them.**
- Side quests = 15 points (5 points per side-quest)

Quantum Trivia Quiz

It’s time!

https://github.com/algolab-iqucodefest/iqucodefest-2026

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